SCIENCE

TRIAL TEST

1. Plant Cells

Multiple Choice Questions (15)

A. General (10 marks)

1. Which part of a plant cell controls all its activities?
   A. Cytoplasm
   B. Cell wall
   C. Nucleus D. Chloroplast
2. The green pigment in plants responsible for photosynthesis is found in the:
   A. Vacuole B. Cell wall C. Chloroplast
   D. Nucleus
3. The cell wall of a plant cell is made up of:
   A. Protein B. Starch C. Cellulose D. Sugar
4. Which of the following is not found in a plant cell?
   A. Cell wall
   B. Chloroplast
   C. Vacuole D. Centriole
5. The liquid part of the cell that holds the organelles is the:
   A. Cytoplasm
   B. Cell wall
   C. Chloroplast
   D. Nucleus
6. The large central vacuole in a plant cell helps to:
   A. Produce energy B. Store water and food C. Control cell activities D. Protect the cell
7. Which part helps the plant cell to make its own food?
   A. Cell wall
   B. Chloroplast
   C. Cytoplasm
   D. Nucleus
8. The outermost layer of a plant cell is called:
   A. Cell wall
   B. Cell membrane C. Cytoplasm D. Nucleus
9. The part that gives the plant cell a definite shape is the:
   A. Cell wall
   B. Chloroplast
   C. Cytoplasm
   D. Nucleus
10. The cell membrane controls:
    A. What enters and leaves the cell B. Cell color C. Cell division D. Growth

B. Scenario-Based (5 marks)

11. A student observed a green part of a leaf under a microscope and saw tiny green structures. What organelles are these?
    A. Nuclei B. Mitochondria C. Chloroplasts D. Ribosomes
12. Kofi crushed a leaf and noticed a watery substance. What part of the cell does this liquid mostly represent?
    A. Cytoplasm
    B. Cell wall
    C. Nucleus D. Vacuole
13. When a plant lacks enough sunlight, which part of its cells cannot function properly?
    A. Nucleus B. Chloroplast
    C. Cell wall
    D. Cytoplasm
14. A leaf cell exposed to heat lost water and shrank. Which structure was affected?
    A. Cell wall
    B. Vacuole C. Cytoplasm D. Chloroplast
15. A student's drawing of a plant cell did not include chloroplasts. What will the cell be unable to do?
    A. Absorb water B. Control activities C. Make food D. Reproduce

Theory Questions (5)

1. Draw and label a typical plant cell (at least 6 parts).
2. State three functions of the cell wall.
3. Mention four differences between plant and animal cells.
4. Explain why plant cells are green in color.
5. What is the function of the vacuole in a plant cell?

2. Animal Cells

Multiple Choice Questions (15)

A. General (10 marks)

1. Which part of an animal cell controls cell activities?
   A. Cytoplasm
   B. Nucleus C.

Cell wall D. Chloroplast

2. The jelly-like part of an animal cell is the:
A. Cytoplasm B. Nucleus C. Vacuole D. Membrane
3. Animal cells do not have:
A. Cell wall B. Cytoplasm C. Nucleus D. Membrane
4. Which structure allows materials to pass in and out of a cell?
A. Cell membrane B. Nucleus C. Vacuole D. Cytoplasm
5. The power house of the cell is the:
A. Chloroplast B. Mitochondrion C. Vacuole D. Ribosome
6. The function of the cytoplasm is to:
A. Protect the cell B. Store food C. Hold organelles D. Make food
7. Which of the following is common to both plant and animal cells?
A. Cell wall B. Chloroplast C. Nucleus D. None
8. The thin layer surrounding the cell is the:
A. Cell membrane B. Cell wall C. Cytoplasm D. Nucleus
9. The part that stores food and water in animal cells is the:
A. Nucleus B. Cytoplasm C. Vacuole D. Ribosome
10. The main difference between plant and animal cells is that animal cells lack:
A. Cell membrane B. Nucleus C. Chloroplast D. Cytoplasm

B. Scenario-Based (5 marks)

11. A scientist looks at a cell without a cell wall. What type of cell is it likely to be?
A. Animal B. Plant C. Bacterial D. Fungal

12. Adwoa observes a round cell without green color under a microscope. What can she conclude?
A. It is an animal cell B. It is a leaf cell C. It contains chloroplasts D. It can make food
13. If a part of the animal cell controlling all activities is damaged, which organelle is affected?
A. Cell membrane B. Nucleus C. Vacuole D. Cytoplasm
14. A student's cheek cell was examined under a microscope. Which part helped him identify it as animal cell?
A. Absence of cell wall B. Presence of chloroplast C. Large vacuole D. Green color
15. If food particles enter an animal cell, which part helps digest them?
A. Lysosome B. Nucleus C. Chloroplast D. Mitochondrion

Theory Questions (5)

1. Draw and label an animal cell.
2. State three functions of the cell membrane.
3. Mention four differences between animal and plant cells.
4. Define a cell.
5. Why do animal cells not have cell walls?

3. Differences and similarities between plant and animal cells

Multiple Choice Questions (15)

A. General (10 marks)

1. Both plant and animal cells contain:
A. Chloroplasts B. Cell walls C. Nuclei D. None
2. Which of these is found only in plant cells?
A. Nucleus B. Cell wall C. Cytoplasm D. Mitochondria
3. Which of the following is not a similarity between plant and animal cells?
A. Both have nuclei B. Both have chloroplasts C. Both have cytoplasm D. Both have membranes
4. The cell wall is absent in:
A. Animal cells B. Plant cells C. Both D. None
5. Both plant and animal cells perform:
A.

Photosynthesis B. Reproduction C. Water transport D. Food storage

6. Chloroplasts are used for:
A. Respiration B. Photosynthesis C. Digestion D. Excretion
7. Which part gives plant cells their box-like shape?
A. Cell membrane B. Cytoplasm C. Cell wall D. Vacuole
8. The main difference in food-making ability is that:
A. Both can make food B. Only animal cells can C. Only plant cells can D. None can
9. Both cells need which part for controlling activities?
A. Nucleus B. Chloroplast C. Cell wall D. Ribosome
10. The large vacuole is more common in:
A. Animal cells B. Plant cells C. Both D. None

B. Scenario-Based (5 marks)

11. When a green leaf cell and cheek cell are compared, which structure appears in only one?
A. Nucleus B. Cytoplasm C. Chloroplast D. Membrane
12. Ama drew a rectangular cell with a thick outer layer and green spots. What cell is it?
A. Animal cell B. Plant cell C. Bacterial cell D. Fungal cell
13. Yaw found two types of cells: one can make food, the other cannot. Which difference is seen?
A. One has nucleus B. One has chloroplast C. One has mitochondria D. One has ribosome
14. After observing a transparent cell without a cell wall, which conclusion can be drawn?
A. It's an animal cell B. It's a plant cell C. It's a bacteria D. It's a fungus
15. A cell shrinks when it loses water. Which part helped maintain water balance?
A. Vacuole B. Chloroplast C. Cytoplasm D. Cell wall

Theory Questions (5)

1. Write 4 differences between plant and animal cells.
2. List 3 similarities between them.
3. Explain why only plant cells contain chloroplasts.
4. Why is a cell called the basic unit of life?
5. Draw a table comparing plant and animal cells (5 points each).

4. Tissues

Multiple Choice Questions (15)

A. General (10 marks)

1. A group of similar cells performing the same function is called a:
A. Tissue B. Organ C. System D. Organism
2. Which of the following is not a type of animal tissue?
A. Muscle B. Epithelial C. Xylem D. Nerve
3. Plant tissues that carry water are called:
A. Phloem B. Xylem C. Cambium D. Cortex
4. Blood is an example of:
A. Muscle tissue B. Connective tissue C. Nervous tissue D. Epithelial tissue
5. The main function of muscle tissue is:
A. Protection B. Movement C. Transport D. Support
6. Plant tissues are grouped into:
A. Transport and support B. Xylem and phloem C. Conducting and supporting D. All of the above
7. Nerve tissues help in:
A. Movement B. Coordination C. Transport D. Protection
8. The xylem carries:
A. Water B. Food C. Oxygen D. Blood
9. The phloem carries:
A. Minerals B. Water C. Manufactured food D. Salts
10. Connective tissues help to:
A. Link body

parts B. Move the body C. Protect organs D. Absorb food

B. Scenario-Based (5 marks)

11. Kofi cut his finger and blood came out. Which tissue is this? A. Connective B. Epithelial C. Nerve D. Muscle

12. A plant stem stands upright because of which tissue? A. Xylem B. Phloem C. Support tissue D. Epidermis
13. When Ama touches a hot object and pulls her hand away, which tissue helped her respond? A. Nerve B. Muscle C. Connective D. Blood
14. A farmer observed that a plant's leaves were dry due to blocked water vessels. Which tissue is affected? A. Xylem B. Phloem C. Cambium D. Cortex
15. The beating of the heart is due to which tissue? A. Connective B. Muscle C. Nerve D. Epithelial

Theory Questions (5)

1. Define tissue.
2. Name two types of plant tissues and their functions.
3. Mention four types of animal tissues and their functions.
4. Explain why blood is a tissue.
5. Describe how xylem and phloem help a plant to survive.

5. Organs

Multiple Choice Questions (15)

A. General (10 marks)

1. An organ is made up of: A. One cell B. Many tissues C. One tissue D. Organ systems
2. The heart is an organ for: A. Digestion B. Circulation C. Breathing D. Reproduction
3. The main organ of the respiratory system is the: A. Stomach B. Lung C. Kidney D. Liver
4. The organ that filters blood is the: A. Heart B. Kidney C. Liver D. Brain
5. Which of the following is a plant organ? A. Xylem B. Leaf C. Phloem D. Pith
6. The organ responsible for food digestion is the: A. Heart B. Stomach C. Brain D. Lung
7. The root functions mainly to: A. Make food B. Absorb water C. Breathe D. Protect
8. The brain controls: A. Digestion B. Circulation C. Body activities D. Breathing
9. Which of these is not an organ? A. Liver B. Heart C. Bone D. Cell
10. In plants, the leaf helps in: A. Respiration B. Photosynthesis C. Absorption D. Reproduction

B. Scenario-Based (5 marks)

11. Kwaku's kidneys failed. What process in his body will stop? A. Digestion B. Waste removal C. Circulation D. Breathing

12. A plant left in the dark for a week began to turn yellow. Which organ was affected? A. Stem B. Leaf C. Root D. Flower
13. When blood stops circulating, which organ may not be working? A. Heart B. Lung C. Kidney D. Liver
14. If a person's stomach produces less digestive juice, which process is affected? A. Breathing B. Digestion C. Thinking D. Excretion
15. A tree was uprooted by wind. Which plant organ was damaged most? A. Leaf B. Root C. Flower D. Stem

Theory Questions (5)

1. Define an organ.
2. Name three organs in the human body and state their functions.

3. Mention three plant organs and their functions.
4. Explain the relationship between tissues and organs.
5. Draw and label a leaf showing its main parts.

6. Farming Systems in Ghana

Multiple Choice Questions (15)

A. General (10 marks)

1. Farming system means:
A. Type of soil B. Method of crop and animal production C. Farm tool D. Rainfall pattern
2. Mixed farming involves:
A. Only crops B. Only animals C. Both crops and animals D. None
3. Crop rotation helps to:
A. Waste land B. Reduce soil fertility C. Maintain soil fertility D. Increase pests
4. Monocropping means:
A. Growing one crop B. Growing two crops C. Growing crops and animals D. Changing crops yearly
5. Shifting cultivation is mainly practiced where:
A. Land is scarce B. Land is abundant C. Land is dry D. There's irrigation
6. Bush fallowing means:
A. Leaving land uncultivated B. Burning land C. Planting one crop D. Watering daily
7. The best farming system for small farmers with animals is:
A. Mixed farming B. Crop rotation C. Monocropping D. Bush fallowing
8. In crop rotation, legumes are planted because they:
A. Use nitrogen B. Fix nitrogen C. Store water D. Prevent weeds
9. Intensive farming means:
A. Using large land B. Using small land with high input C. Leaving land fallow D. Moving from place to place
10. Extensive farming means:
A. Large area, low input B. Small area, high input C. Irrigated land D. Urban farming

B. Scenario-Based (5 marks)

11. A farmer grows maize this year and beans next year on the same land. What system is this?
A. Mixed farming B. Crop rotation C. Monocropping D. Shifting cultivation

12. Efua rears goats and grows cassava on the same farm. Which farming system does she practice?
A. Mixed farming B. Monocropping C. Crop rotation D. Bush fallowing
13. Kofi left his farmland unused for 3 years to regain fertility. What system is that?
A. Shifting cultivation B. Bush fallowing C. Mixed farming D. Crop rotation
14. A farmer uses fertilizers and irrigation on a small plot of land. What is this called?
A. Extensive farming B. Intensive farming C. Bush fallowing D. Mixed cropping
15. A cocoa farmer plants only cocoa for many years. What system is that?
A. Monocropping B. Crop rotation C. Mixed farming D. Shifting cultivation

Theory Questions (5)

1. Define a farming system.
2. Explain the advantages of mixed farming.
3. List three differences between intensive and extensive farming.
4. Describe two problems of shifting cultivation.
5. State three ways crop rotation improves soil fertility.